IMEN574 / IDSC574 · PROGRAMMING FOR AX

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Training and Generalization

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Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Training and Generalization

01 **Generalization Evidence**

Read the data boundary

02 **Optimization Diagnostics**

Control update behavior

03 **Regularization and Experiments**

Test controlled changes

01 · Generalization Evidence

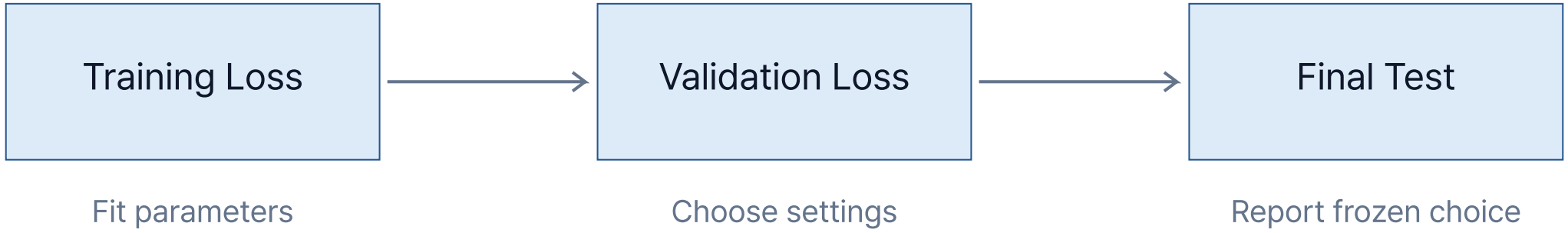
01 **Generalization Evidence**
Read the data boundary

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Training Evidence

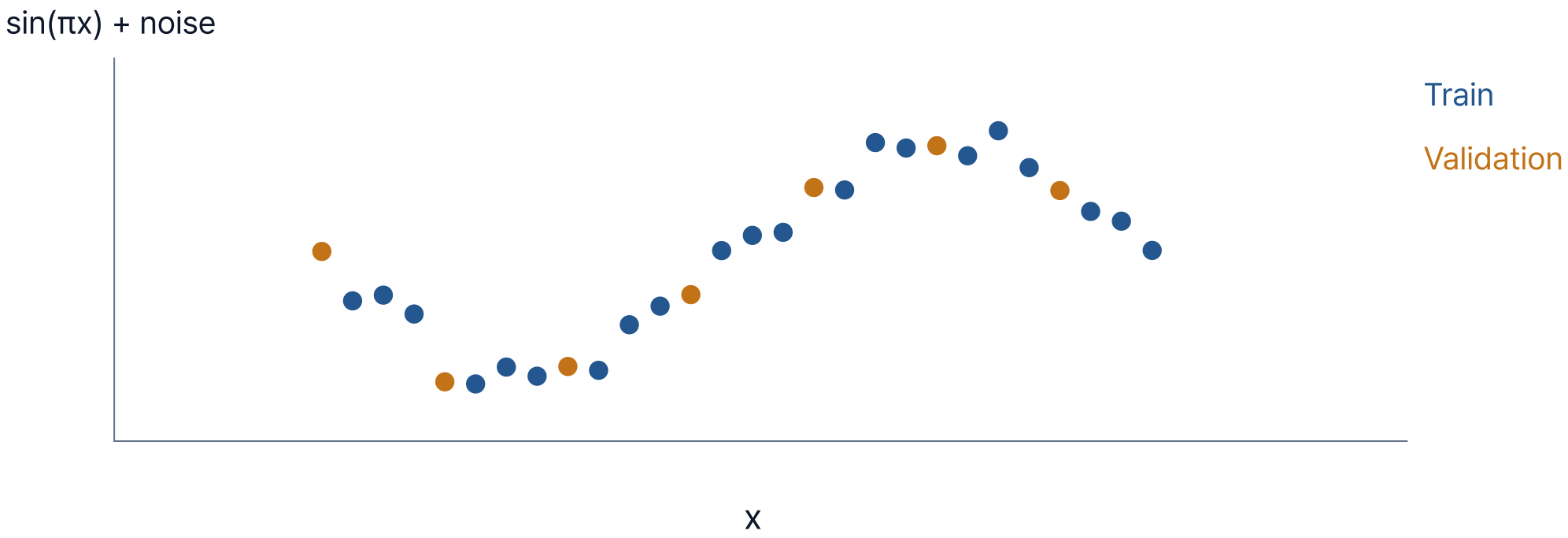
Separate optimization progress from generalization evidence



Each dataset has a different decision role

A Controlled Noisy Dataset

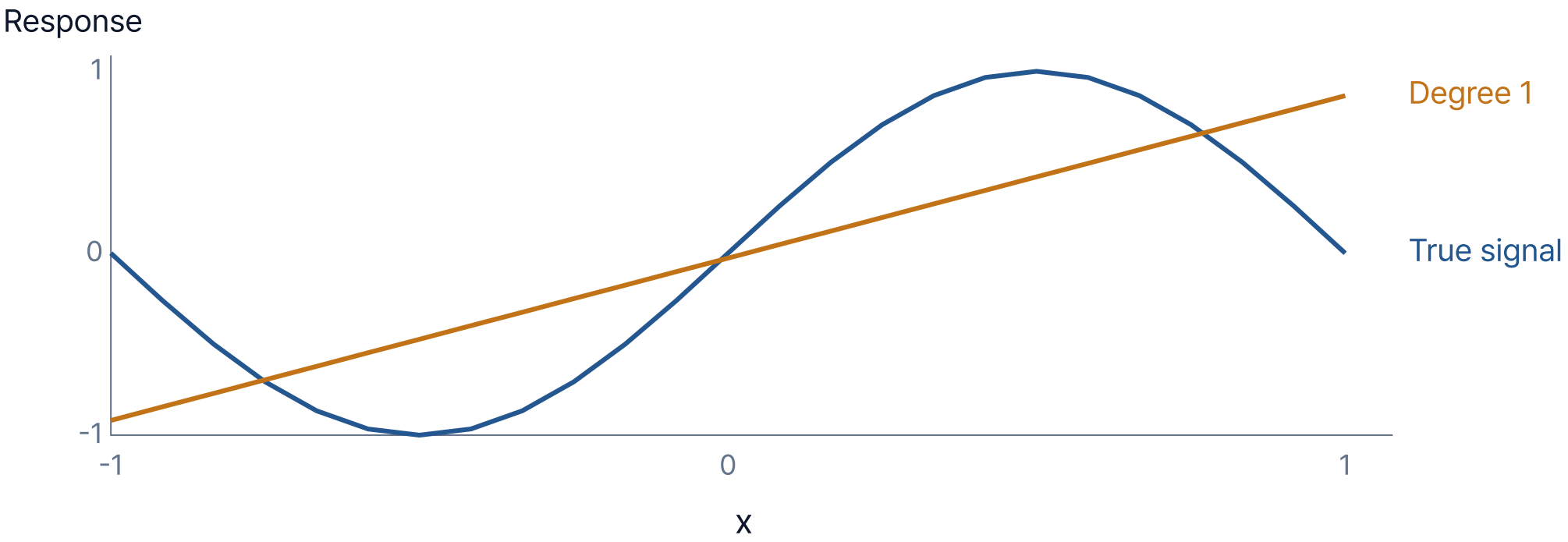
Use known signal plus noise to expose fitting behavior



The held-out points were separated before model comparison

Underfitting Evidence

A simple family can miss systematic structure



The linear family cannot follow both bends

Capacity and Training Fit

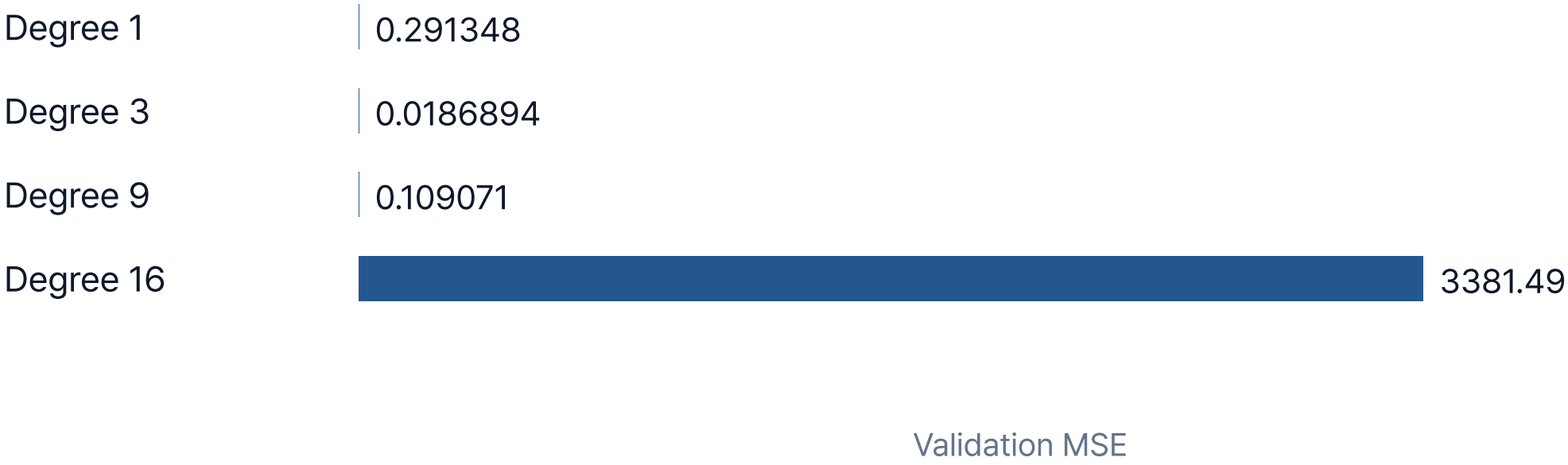
More flexible models can reduce observed training error



Training error decreases as polynomial capacity increases

Capacity and Validation

Better training fit can coexist with worse held-out error



The held-out loss distinguishes flexible fits

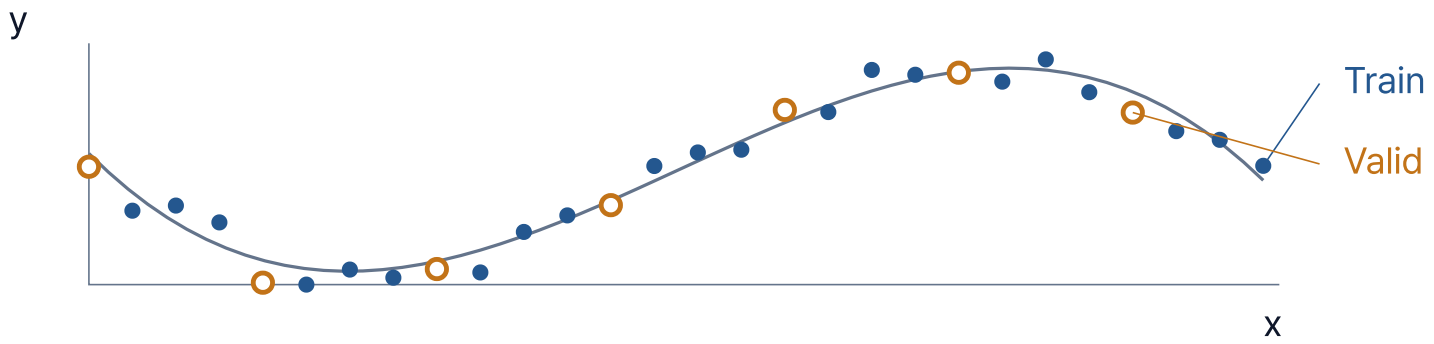
Demo: Capacity on a Fixed Validation Split

Does degree nine improve validation? Can ridge reduce the high-degree error?

Ordered x, y rows

```
-1.000000000,0.045707562;
-0.925925926,-0.386613487;
-0.851851852,-0.336231501;
```

Polynomial degree 3 Ridge λ 0



Actual polynomial fit | degree 3, ridge 0; train 21, validation 7
Training MSE 0.01814 | validation MSE 0.01869
Every fourth input row is held out before fitting; the test set is not used here

Actual least-squares fit on the notebook's 28 noisy observations

Answer: Capacity and Held-Out Error

The higher-degree fit has lower training error and worse validation error

Degree	Training MSE	Validation MSE
3	0.018137888	0.018689447
9	0.008022128	0.109070750

A better training fit can be a worse held-out predictor

Caveat: Choosing on Validation

A held-out set becomes selection evidence when used repeatedly

Operation	Allowed Role
Fit coefficients	Training rows
Choose degree and ridge	Validation rows
Report final choice	Untouched test rows

The displayed validation score is not an untouched final test

Early Stopping

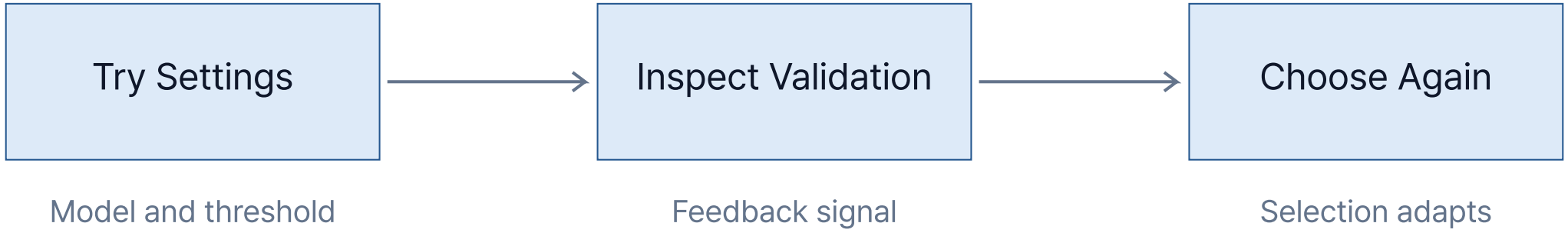
Select a saved state before the full training budget ends

Epoch	Validation Loss
5	0.45
10	0.31
15	0.25
20	0.29

The selected state is epoch fifteen in this example

Caveat: Validation Reuse

Repeated choices can adapt to the validation set



The validation set becomes part of the selection procedure

Data Leakage Diagnosis

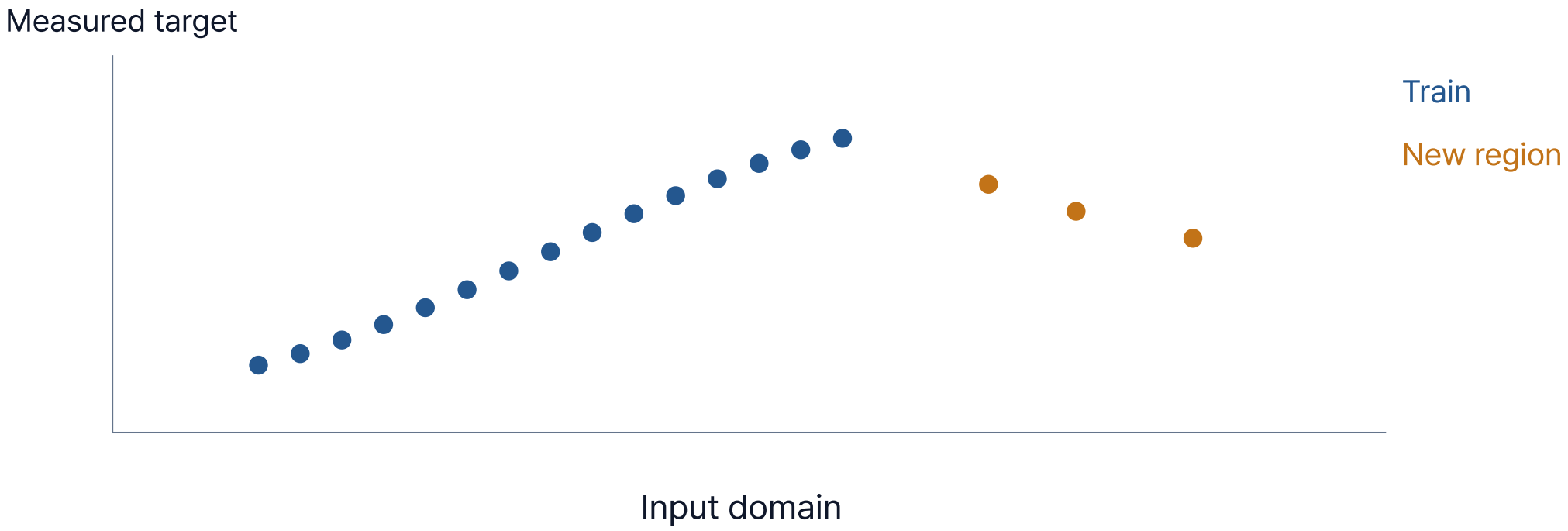
A suspiciously strong score warrants a boundary audit

Leak	Example
Target timing	Future failure status
Entity overlap	Same machine in both splits
Transform fit	Scale on all rows
Selection	Tune using test labels

The split boundary matters before optimizer sophistication

Distribution Shift Evidence

Compare deployment inputs with the training coverage



New operating regions require evidence beyond interpolation

Error Slices

Averages can hide a small failing subgroup

	Cases	Mean Loss
Common	90	0.1
Rare	10	1.0
Overall	100	0.19

A low overall loss can hide ten difficult cases

02 · Optimization Diagnostics

01 Generalization Evidence

Read the data boundary

02 **Optimization Diagnostics**

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The Learning Rate

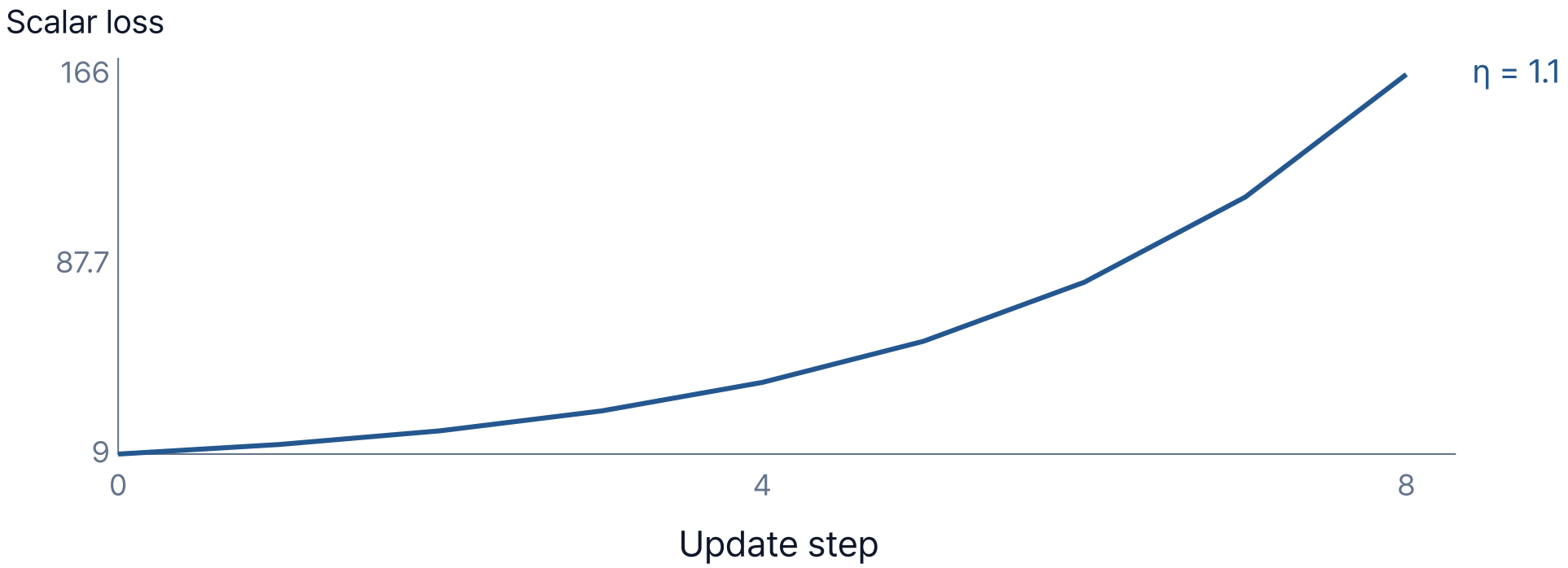
The step size controls how far each gradient moves



Both rates contract this quadratic error

Caveat: Divergent Updates

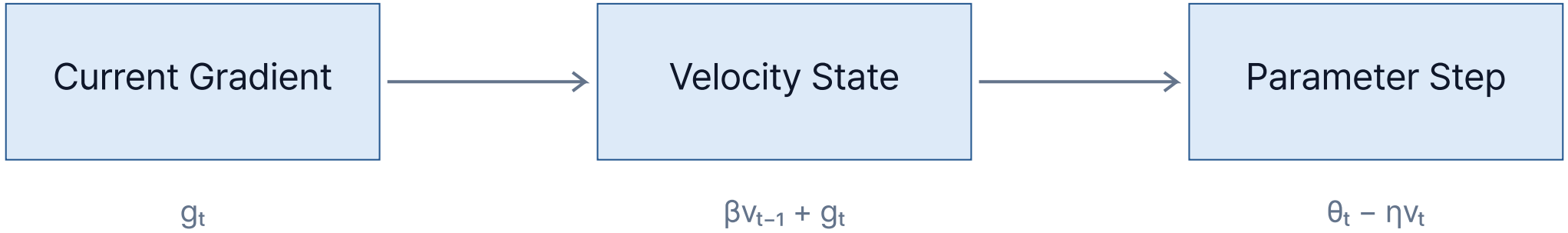
An accurate gradient can produce growing loss



The error grows when the multiplier magnitude exceeds one

Momentum State

Retain a moving direction across gradient steps



The optimizer carries state beyond the current gradient

Adam Moment Estimates

Track the first and second moments before scaling the update

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\hat{m}_t = m_t / (1 - \beta_1^t) \quad \hat{v}_t = v_t / (1 - \beta_2^t)$$

$$\theta_t = \theta_{t-1} - \eta \, \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)$$

Both moment estimates are optimizer state

https://docs.pytorch.org/docs/stable/generated/torch.optim.Adam.html

AdamW Weight Decay

Decay parameters separately from the moment-based gradient update

Gradient path: $g_t \rightarrow m_t, v_t \rightarrow$ adaptive update

Decay path: $\theta \leftarrow (1 - \eta\lambda)\theta$

$$\theta_t = (1 - \eta\lambda)\theta_{t-1} - \eta\, \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)$$

Decoupled decay does not enter the moment estimates

https://docs.pytorch.org/docs/stable/generated/torch.optim.AdamW.html

A Fixed Optimization Experiment

Keep data and feature definitions unchanged across optimizer settings

Component	Value
Training / validation	21 / 7 fixed rows
Model	Nine polynomial features + bias
Initialization	All parameters zero
Dropout	Inverted feature masks, seed 7
Evaluation	Unmasked features

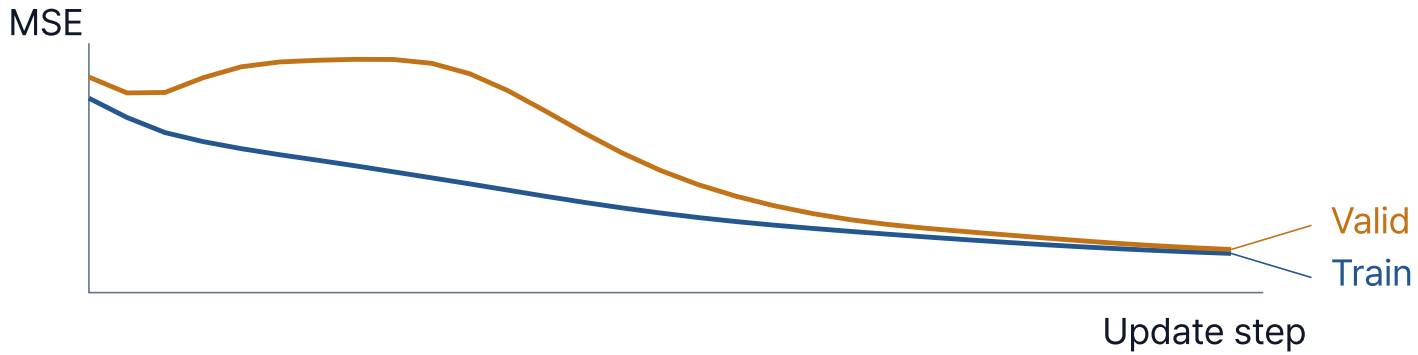
Each run starts from the same explicit input and initial state

Demo: Optimizer and Regularization Updates

What changes after thirty versus 120 updates? Does dropout or decay improve this fixed validation split?

Optimizer AdamW Step η 0.05 Updates 30

Decay λ 0 Feature dropout 0



Explicit degree-9 feature model, not PyTorch | adamw, steps 30, η 0.05
Eval-mode train MSE 0.1006, validation MSE 0.1107; $\|w\|_2$ 1.688
w_1 1.183; m_1 -0.1562, v_1 0.001852 | eval has no mask

Explicit SGD / AdamW and feature dropout; no PyTorch runtime

Answer: The Same AdamW Run

Only the update budget changes between the two rows

Updates	Training MSE	Validation MSE
30	0.100640456	0.110746081
120	0.033025849	0.064536041

The optimizer state and losses come from executed parameter updates

Caveat: Dropout and Evaluation

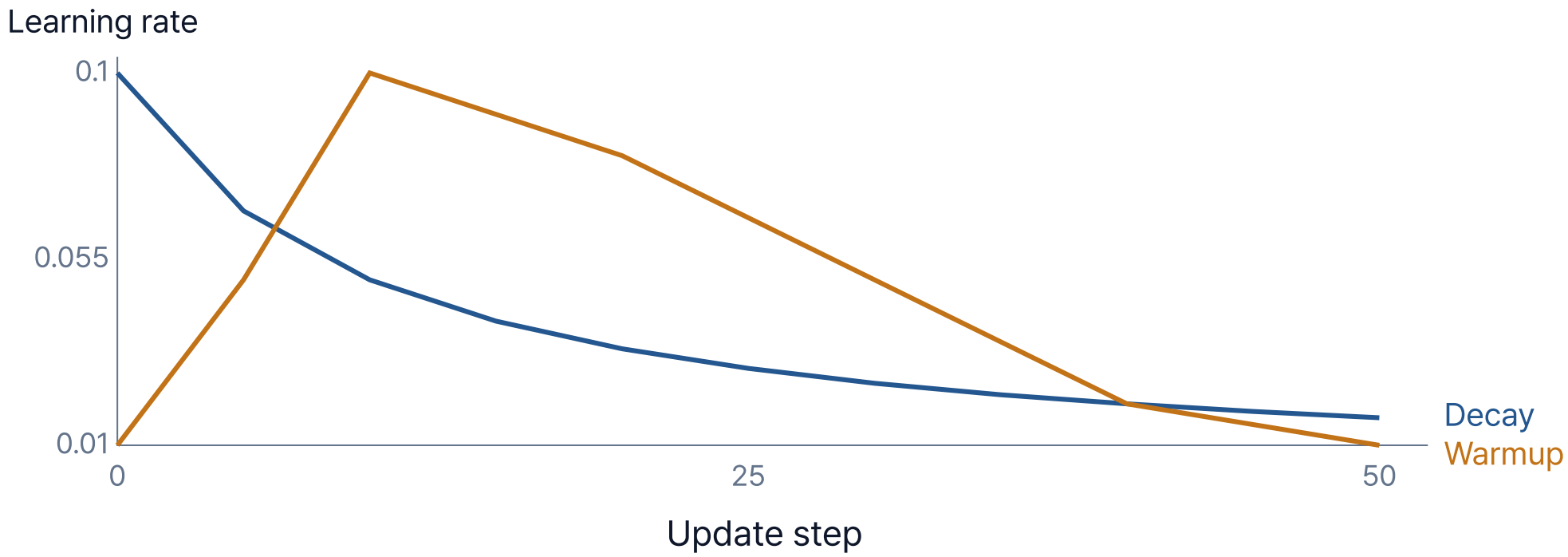
Training masks must not randomly change the validation computation

Stage	Feature Mask	Parameter Updates
Training	Sampled and rescaled	Yes
Validation	All features restored	No

Different masks are training inputs rather than validation evidence

Learning Rate Schedules

Change the update scale during a run



Warmup increases the scale before the later decay

Initialization Scale

Choose weight scale to avoid immediate saturation or explosion



Illustrative activation variance

Initial scale changes the magnitude of forward signals

Gradient Clipping

Limit the size of an exceptional update signal

	g_1	g_2	$\|g\|$
Before	3	4	5
Norm Clipped	1.2	1.6	2

$$g_{clipped} = g \times \min(1, c / \|g\|)$$

Clipping to norm two preserves the gradient direction

Normalization Axes

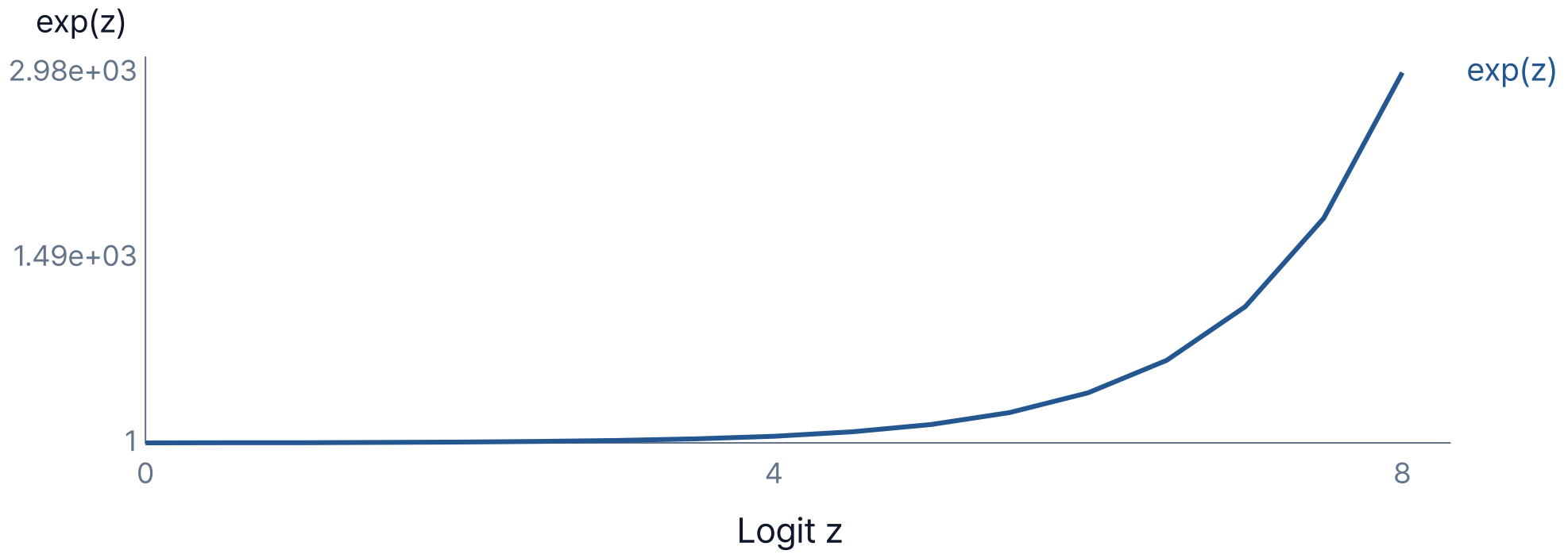
Batch and layer normalization aggregate different dimensions

Statistics Axis	Method
Across observations	BatchNorm
Within one example	LayerNorm

The statistic axis determines which examples interact

Numerical Stability

Preserve the same objective using stable arithmetic



Exponentials grow rapidly even for moderate positive logits

03 · Regularization and Experiments

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Regularization Mechanisms

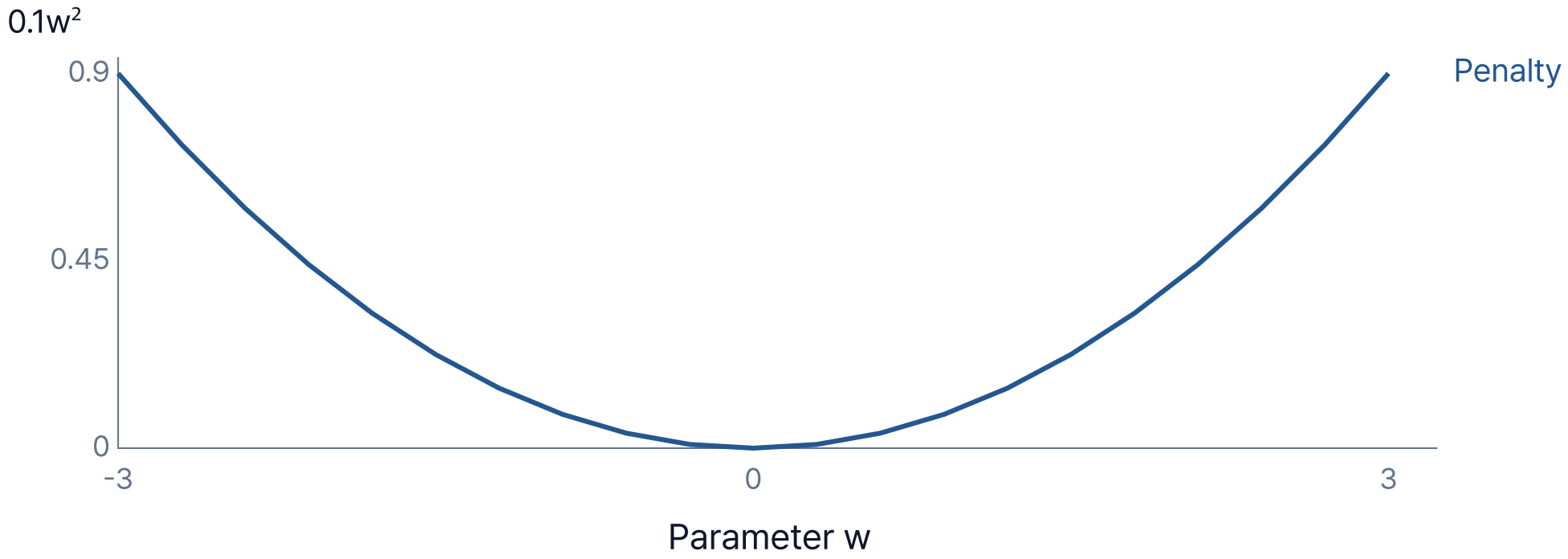
Different controls restrict different failure modes

Control	Primary Effect
Weight decay	Penalize parameter magnitude
Dropout	Randomize hidden activations
Early stopping	Limit fitting duration
Augmentation	Expand chosen input variations

Regularization changes either the objective or the learning experience

Weight Decay and Model Scale

A magnitude penalty changes the optimization tradeoff



Large parameter magnitudes pay a larger penalty

Dropout Masks

Training uses random masks with a scale correction

	h_1	h_2	h_3	h_4
Activation	2	4	6	8
Keep Mask	0	1	0	1
Scaled Output	0	8	0	16

At keep probability one half, retained values double during training

Augmentation Validity

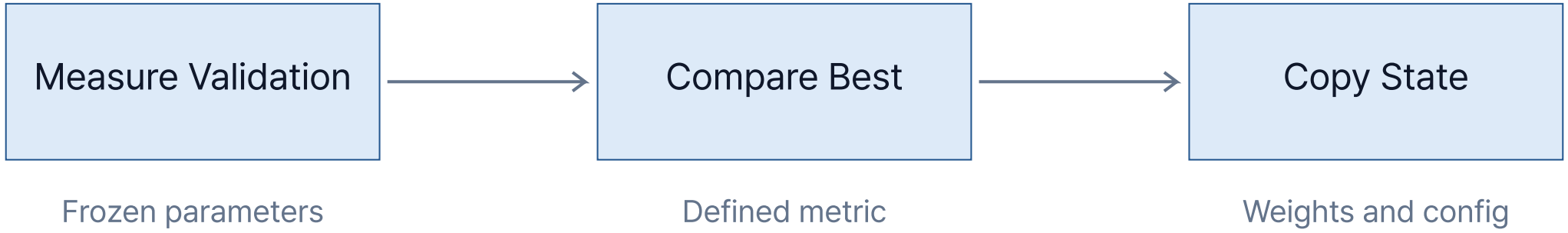
Preserve the task label under the chosen input change

Change	Potentially Valid	Potential Failure
Small image crop	Object retained	Target removed
Text substitution	Meaning retained	Negation changed
Time shift	Event unchanged	Label timing broken

An augmentation is valid only under a task-specific invariance

Checkpoint Selection

Save the best validation state with its configuration



The retained state should correspond to the selected score

Multiple Seeds

Separate an apparent gain from run-to-run variation

	Seed 1	Seed 2	Seed 3
Baseline	0.31	0.29	0.32
Candidate	0.3	0.28	0.31

Compare paired runs before treating a small gap as decisive

Ablation as a Test

Change one component while holding the protocol fixed

Run	Decay	Augmentation
Base	Off	Off
+ Decay	On	Off
+ Augment	Off	On

Separate changes reveal which intervention explains a result

Training Failure Triage

Use the observed failure to choose the next check

Observation	Next Check
Tiny batch cannot fit	Code, loss, gradient
Train good, valid poor	Coverage and capacity
NaN appears	First unstable operation
All classes identical	Targets and imbalance

A diagnostic action follows the nearest visible symptom

Experiment Records

Retain enough evidence to repeat a selected run

Record	Example
Data boundary	Row IDs and split rule
Configuration	Model, optimizer, seed
Evidence	Losses and error slices
Artifact	Selected checkpoint and version

The result includes its data boundary and selected state

Generalization Decision

Use a frozen evaluation to justify the final choice



A useful training run ends with tested deployment assumptions

Takeaway: Training and Generalization

01 Distinguish fitting from generalization evidence

02 Diagnose gradients and numerical scale before adding complexity

03 Retain a selected state and an honest evaluation boundary